

Why the Icosahedron

Three, Five, and Non-Commutation

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Why the Icosahedron: Three, Five, and Non-Commutation

Abstract

FFGFT is a $\mathbb{Z}_3$ theory from the ground up. Yet the lepton masses contain the golden ratio ϕ , and ϕ is the number of five. This document explains why the icosahedron is the bridge: its rotation group A_5 is the smallest finite group in which a three and a five coexist without commuting. From this non-commutation arises a mixing of the $\mathbb{Z}_3$ modes under the five-fold rotation, and the mixing has one rational component $2/9$ and two ϕ -bearing components. The rational one is the Koide phase (Doc. 367), the ϕ -bearing ones are the skeleton of the mass ratios (Doc. 368). A single matrix element of this mixing carries the entire Koide formula. The document computes nothing new; it narrates what Docs. 285, 293, 367 and 368 mean together.

Notation

Epistemic markers:

[B]	algebraically proved (check script in the cited document)
[K]	numerically confirmed
[S]	structurally observed, derivation open
[E]	established external mathematics

Symbol	Meaning
$\mathbb{Z}_3, C_3$	cyclic group of order 3; C_3 : permutation $(x, y, z) \mapsto (y, z, x)$
v_0, v_1, v_2	Fourier modes, eigenvectors of C_3
ω	$e^{2\pi i/3}$
A_5	icosahedral group, 60 elements
R_5	rotation by 72° about the icosahedron vertex $(0, 1, \phi)$
ϕ	golden ratio $(1 + \sqrt{5})/2$
p_0, p_1, p_2	redistribution weights of the electron mode under R_5 (Doc. 293)
A	$\langle v_0 R_5 v_e \rangle, A = \sqrt{2}/3$ (Doc. 367)
D_4	lattice on T^4 ; $ \text{Aut}(D_4) = 1152$ (Docs. 314, 364)
$\text{GF}(3^k)$	finite field of characteristic 3; $\text{GF}(9) \ni \phi, \text{GF}(81) \ni \zeta_5$

1 The Observation

There are three charged leptons. Their masses stand in the ratio $m_e : m_\mu : m_\tau \approx 1 : 206.8 : 3477.4$, and since 1981 this ratio has been known to obey a strikingly simple formula, the Koide formula:

$$\sqrt{m_k} = \sqrt{M} \left(1 + \sqrt{2} \cos \left(\theta + \frac{2\pi k}{3} \right) \right), \quad k = 0, 1, 2.$$

The scale M is a unit – it only says how large the masses are overall. The phase θ is the actual information: it determines the ratios. Empirically $\theta = 2/9$ to 10^{-7} (Doc. 292). Why $2/9$? Why not $1/4$, not 0.3 ? And why the coefficient $\sqrt{2}$?

The answer leads through a geometric object that at first has nothing to do with leptons: the icosahedron. The rest of this document explains why.

2 Where FFGFT Comes From: The Three

FFGFT begins with a $\mathbb{Z}_3$. Three generations, triality, the circulant as mass operator. The mode space is $\mathbb{C}^3$; its natural basis is the Fourier modes

$$v_k = \frac{1}{\sqrt{3}} (1, \omega^k, \omega^{2k}), \quad k = 0, 1, 2.$$

These are the eigenvectors of the cyclic permutation C_3 . The mode v_0 is the symmetric one; v_1 and v_2 carry the phases ω and ω^2 . The electron sits in v_1 , the muon in v_2 , the tau in v_0 – this is the assignment carried by the Koide index k .

What this structure alone does *not* deliver is a number. The $\mathbb{Z}_3$ says there are three modes and how they transform under C_3 . It says nothing about masses. The Koide formula needs two inputs – the coefficient $\sqrt{2}$ and the phase $\theta = 2/9$ – and neither is contained in the $\mathbb{Z}_3$.

3 Where ϕ Comes From: The Five

The golden ratio is the number of the pentagon: $\cos 72^\circ = (\phi - 1)/2$, and the diagonals of a regular pentagon stand to its sides in the ratio ϕ [E]. There is no five in the $\mathbb{Z}_3$ structure. If ϕ appears in the lepton masses – Doc. 368 shows $p_1/p_2 = \phi^8$ exactly [B] – a five must come from somewhere.

4 Why the Icosahedron: Where Three and Five Meet

The icosahedron has 20 triangular faces and 12 vertices; at each vertex five triangles meet. Its rotation group A_5 has 60 elements: the identity, 15 rotations by 180° (edge midpoints), 20 rotations by 120° (face centres), 24 rotations by 72° or 144° (vertices) [E]. Three and five are both present.

That is not trivial. The cyclic group $\mathbb{Z}_{15}$ also contains elements of order 3 and 5 – but there they commute. In A_5 they *do not commute*: a rotation by 120° followed by one of 72° is not the same as the reverse order. A_5 is the smallest group in which this is so.

Concretely: in standard position the vertices of the icosahedron are the cyclic permutations of $(0, \pm 1, \pm \phi)$. The 3-fold axes pass through face centres; one of them is $(1, 1, 1)$, and the rotation by 120° about this axis is *exactly* the permutation C_3 of FFGFT. The $\mathbb{Z}_3$ of FFGFT is, without reinterpretation, a subgroup of the icosahedral group (Doc. 285). The 5-fold axes pass through vertices; one vertex is $(0, 1, \phi)$, and the rotation R_5 by 72° about this axis is the five-fold rotation used in Doc. 293.

5 Where the Icosahedron Lives: Unrolled, Not Compact

Up to here it sounds as if the icosahedron were simply *the* structure of FFGFT. It is not – and the corpus is very precise about this (Docs. 285, 314, 364). FFGFT has two pictures connected by a lossless coordinate transformation, and the icosahedron lives in only one of them.

Compact: the torus T^4 with the D_4 lattice. This is the original. The four-torus carries the D_4 lattice, and its triality is the $\mathbb{Z}_3$ of FFGFT – as a lattice operation, as a circulant on the mode orbits (Doc. 314). The winding numbers of the T0 ladder live here. The lattice knows the three. It knows *no* five: $|\text{Aut}(D_4)| = 1152 = 2^7 \cdot 3^2$, and $5 \nmid 1152$. No icosahedron acts on a D_4 lattice (Doc. 364, incompatibility theorem) [B].

Unrolled: $\mathbb{R}^3 \times S^1$. Observed space arises when one circle of the torus becomes time through the measurement process: $T^4 \rightarrow \mathbb{R}^3 \times S^1$ (Doc. 285). On *this* $\mathbb{R}^3$, A_5 acts as rotation group. The $C_3 \subset A_5$ is the same three as the triality in the compact picture – the rotation by 120° about $(1, 1, 1)$. But the five exists only here. The icosahedron is the image of the torus geometry in unrolled space, not the torus itself.

How the five nevertheless enters the field. Doc. 364 resolves the apparent tension: what matters is *as what* a symmetry enters – as an automorphism of lattice vectors or as a phase. The 72° rotation has characteristic polynomial $(x-1)(x^2-(\phi-1)x+1)$; its trace is ϕ . Modulo 3, ϕ lies in $\text{GF}(9)$ and is primitive there ($\phi^4 = -1$, order 8). The roots of the quadratic factor lie not in $\text{GF}(9)$ but in $\text{GF}(81)$, and have order 5 there – because $5 \mid 80 = |\text{GF}(81)^*|$ [B]. In the lattice, the 72° rotation is a vector automorphism, and $5 \nmid 1152$. In the field, the same rotation is only

its eigenvalue ζ_5 , and $5 \mid 80$. Geometric incompatibility becomes arithmetic compatibility as soon as one passes from vectors to phases. Doc. 364 verbatim: this is the mechanism behind Doc. 293 – $\theta = 2/9$ arises from R_5 as a phase, not as a lattice operation.

Why this is forced. In characteristic 3, $x^3 - 1 = (x - 1)^3$ **[E]**: in no $\text{GF}(3^k)$ is there a primitive cube root of unity. The three *cannot* be a phase and must be lattice. The five *cannot* be lattice ($5 \nmid 1152$) and must be phase. The prime 3 distributes the two symmetries onto two complementary channels – not a choice, but a constraint. The extension $\text{GF}(9) \rightarrow \text{GF}(81)$ that carries ζ_5 and ϕ is precisely the doubling $6 = 3 \oplus 3'$ of Doc. 285.

What this means for the narrative. When non-commutation of three and five is spoken of below, this is meant: the three as lattice operation in the compact picture, the five as phase in the unrolled one. They do not meet in geometry – there they are incompatible – but in the field $\text{GF}(81)$. The icosahedron is the visible form of this meeting.

6 What Non-Commutation Produces

Because R_5 does not commute with C_3 , R_5 is *not diagonal* in the mode basis v_0, v_1, v_2 . Applying R_5 to the electron mode v_1 , the result does not land back in v_1 but is distributed over all three modes. The moduli squared of the projections are the redistribution weights (Doc. 293) **[B]**:

$$p_0 = |\langle v_0 | R_5 | v_1 \rangle|^2 = \frac{2}{9}, \quad p_1 = |\langle v_1 | R_5 | v_1 \rangle|^2 = \frac{2 + 3\phi}{9}, \quad p_2 = |\langle v_2 | R_5 | v_1 \rangle|^2 = \frac{5 - 3\phi}{9}.$$

One sees at once: p_0 is rational, p_1 and p_2 contain ϕ . That is structure, not accident. The projection onto the symmetric mode v_0 is blind to the five-fold structure – it only counts how much of the $\mathbb{Z}_3$ mixing arrives, and that is $2/9 = 2/3^2$. The projections onto v_1 and v_2 see the five and therefore carry ϕ .

It becomes even clearer at the symmetric mode itself. What R_5 makes of v_0 :

$$\left(|\langle v_0 | R_5 | v_0 \rangle|^2, |\langle v_0 | R_5 | v_1 \rangle|^2, |\langle v_0 | R_5 | v_2 \rangle|^2 \right) = \left(\frac{5}{9}, \frac{2}{9}, \frac{2}{9} \right) \quad \mathbf{[K]}$$

The complete matrix of moduli squared **[K]**:

$ \langle v_j R_5 v_k \rangle ^2$	$k = 0$	$k = 1$	$k = 2$
$j = 0$	5/9	2/9	2/9
$j = 1$	2/9	$(2 + 3\phi)/9$	$(5 - 3\phi)/9$
$j = 2$	2/9	$(5 - 3\phi)/9$	$(2 + 3\phi)/9$

Two things are immediately visible. Wherever v_0 is involved, the entry is $2/9$ or $5/9$ – rational, without ϕ . And ϕ appears only between v_1 and v_2 , the two modes that see the five. Every row and every column sums to 1.

The amplitudes of the first row are $(\sqrt{5}, \sqrt{2}, \sqrt{2})/3$. The five-fold rotation leaves $\sqrt{5}/3$ of the symmetric mode in itself and passes $\sqrt{2}/3$ to each of the two non-trivial modes. And $5 + 2 + 2 = 9 = 3^2$. This is the one row in which the five of the icosahedron and the three of FFGFT stand side by side. The trace of R_5 is $1 + 2 \cos 72^\circ = \phi$ **[E]** – the five is in the operator, the three in the basis, and the product of both is the row $(5, 2, 2)/9$.

7 Doc. 367: Why $\theta = p_0$, and How the Whole Koide Formula Follows

Doc. 293 had noted that $p_0 = 2/9$ coincides with the Koide angle $\theta = 2/9$, and left the question of the reason open. Doc. 367 gives the answer: both are the same quotient $2/3^2$, read in different roles. Numerator: the non-trivial $\mathbb{Z}_3$ modes. Denominator: the order squared, the dimension of the operator space $\mathbb{C}^3 \otimes \mathbb{C}^3$.

The decisive step comes next. The single matrix element $A = \langle v_0 | R_5 v_e \rangle$ has modulus $\sqrt{2}/3$ and modulus squared $2/9$. The Koide formula

$$a_k = 1 + \sqrt{2} \cos\left(\theta + \frac{2\pi k}{3}\right)$$

needs exactly these two numbers: $\sqrt{2}$ as coefficient, $2/9$ as phase. With $3|A| = \sqrt{2}$ and $|A|^2 = 2/9$ this becomes **[B]**:

$$a_k = 1 + 3|A| \cos\left(|A|^2 + \frac{2\pi k}{3}\right).$$

No free parameter. The modulus gives the amplitude, the modulus squared the phase – Born-rule structure, but here not as interpretation, rather as algebra: the same complex number, once as $|A|$, once as $|A|^2$. The Koide formula is thereby fully derived from the question: how much of the five-fold rotation lands in the symmetric $\mathbb{Z}_3$ mode?

8 Doc. 368: The ϕ -Skeleton

The two other weights carry the five. Their ratios are pure ϕ -powers **[B]**:

$$\frac{p_1}{p_2} = \phi^8, \quad \frac{p_0}{p_2} = 2\phi^4, \quad 3\sqrt{p_j} \in \{\sqrt{2}, \phi^2, \phi^{-2}\}.$$

And $\phi^8 \approx 46.98$ is, up to a Galois factor, the ratio m_τ/m_e : $74\phi^8$ hits to 0.03 %, with ξ correction to 0.003 % **[K]**. That is the second role of the icosahedron: p_0 supplies the phase, p_1 and p_2 supply the skeleton of the mass ratios. Both from the same five-fold rotation, both without ϕ having been inserted anywhere.

9 Doc. 293: Why It Is the Icosahedron and Nothing Else

Doc. 293 tested whether $2/9$ depends on the axis. Two hundred random axes with angle 72° : none yields $2/9$. Angle scan at fixed icosahedral axis: only 72° yields $2/9$. Order series – rotations of order 2, 3, 4, 5, 6 about icosahedral axes: only order 5 yields $2/9$ **[K]**. The number is icosahedron-specific. It arises because the axis $(0, 1, \phi)$ and the angle 72° together stand to the $(1, 1, 1)$ axis of the $\mathbb{Z}_3$ exactly as the icosahedral geometry prescribes.

10 What the Icosahedral $\mathbb{Z}_3$ Is – and What It Is Not

A comparison sharpens the concept. In string theory there are orbifolds $T^6/\mathbb{Z}_3$ whose $\mathbb{Z}_3$ acts as $\omega \cdot \mathbb{1}$ on $\mathbb{C}^3$ – all three coordinates receive the same phase. This $\mathbb{Z}_3$ is *central*: it commutes with every matrix. The icosahedral $\mathbb{Z}_3$ is C_3 with eigenvalues $(1, \omega, \omega^2)$, trace zero, *not central*.

Both structures count $2/3^2$ – the orbifold CFT via the sum of twist-field weights $1/9 + 1/9$ **[E]** (Doc. 367, external parallel). But only the icosahedral $\mathbb{Z}_3$ can be *mixed* by R_5 . For the central one, $\langle v_0 | R_5 | v_1 \rangle$ would vanish, because R_5 would then commute with $\mathbb{Z}_3$ and

preserve sectors. The non-commutation that produced the number above is exactly what distinguishes the icosahedral structure from an orbifold. A_5 has trivial centre; no element of A_5 can become $\omega \cdot 1$ in a faithful representation. The two $\mathbb{Z}_3$'s are different objects that happen – or do not happen – to share the same counting quantity.

11 What Remains Open

Two things [S].

First, the Galois factors. ϕ^8 is exact, but m_τ/m_e is $74\phi^8$, and $74 = 2 \cdot 37$ with $37 \mid |\text{GF}(3^{18})^*|$ comes from the number theory of the fields $\text{GF}(3^n)$ (Doc. 338), not from icosahedral geometry. Whether both share a common root is not known.

Second, the fractal correction. The step from percent to 10^{-3} level is, in all routes (Doc. 369), the same number in different guise – measured, not derived. The Koide route has it hidden in $\cos(2/9)$ and is therefore precise; the algebraic routes must add it back.

12 The Short Version

The icosahedron is the place where three and five do not get along. FFGFT brings the three; the icosahedron is the smallest structure that adds a five without the five commuting with the three. From this non-commutation arises a mixing, and the mixing has one rational component – $2/9$ – and two ϕ -components. The rational one is the Koide phase. The ϕ -components are the skeleton of the masses. A single matrix element of this mixing carries the entire Koide formula, parameter-free.